Parul® University
Vadodara, Gujarat

NAAC GRADE A++

Information and
Communication Technology

Unit 3 : Machine Learning
Artificial Intelligence & Data Science

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Content

1. Machine Learning Introduction
2. Supervised vs. Unsupervised Learning.
3. Implementing ML models using scikit-learn,
4.Modeln evaluation
5. Handling Missing Values
6. overfitting
7. hyperparameter tuning

Machine Learning Overview

- Enables systems to learn patterns from data without explicit programming
- Three main paradigms: Supervised, Unsupervised, Reinforcement Learning
- Focus today: Supervised & Unsupervised Learning
- Practical implementation using scikit-learn library
- Critical considerations: Model evaluation, overfitting, hyperparameter tuning

Supervised Learning

Learning from labeled training data with input-output pairs

- **Classification**

Predicts categorical output (classes/categories).

Examples: Email spam, image recognition

- Regression

Predicts continuous numerical output

Examples: Price prediction, temperature forecast

Unsupervised Learning

Learning patterns from unlabeled data without predefined outputs

Clustering

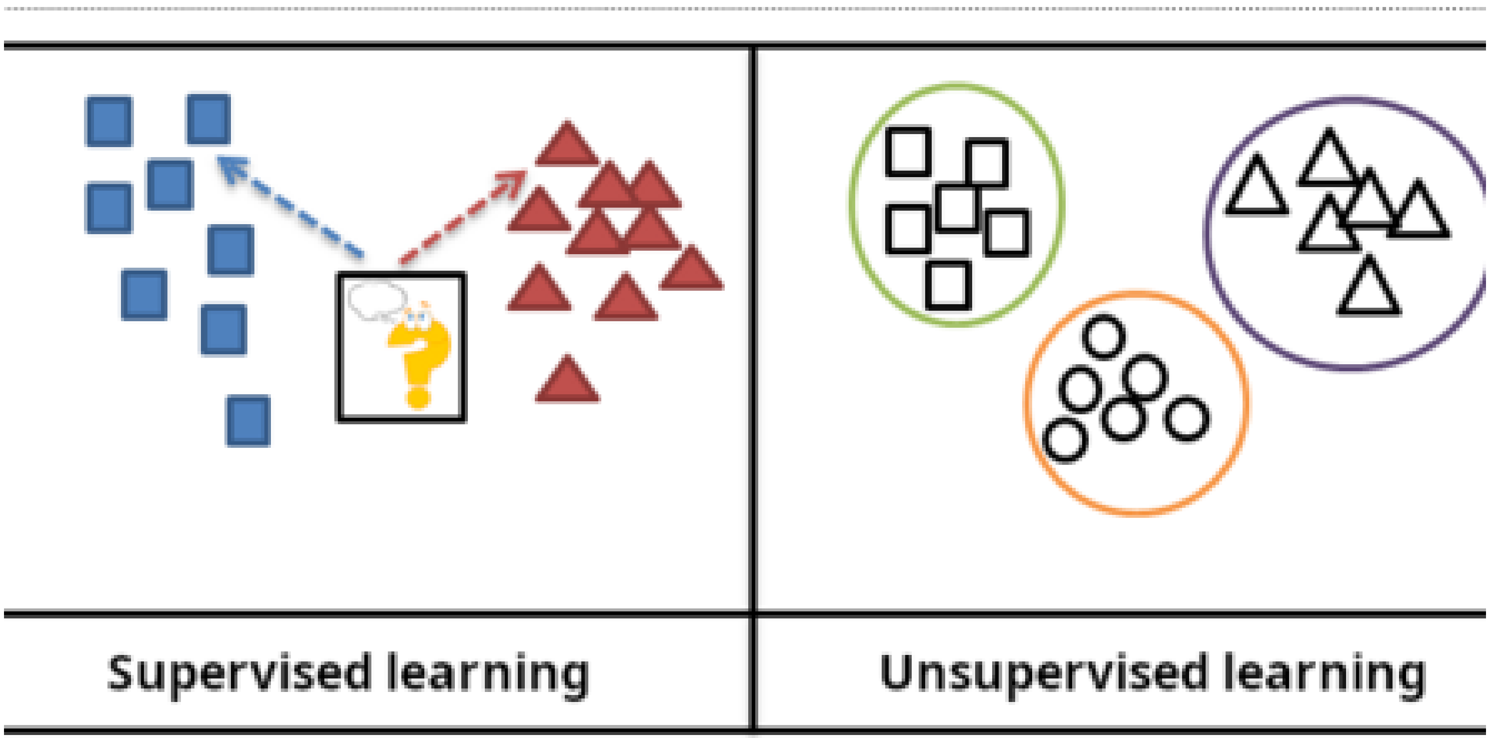
Groups similar data points together

Examples: K-Means, DBSCAN

Dimensionality Reduction

- **Reduces feature space while preserving information**
- **Examples: PCA, t-SNE**

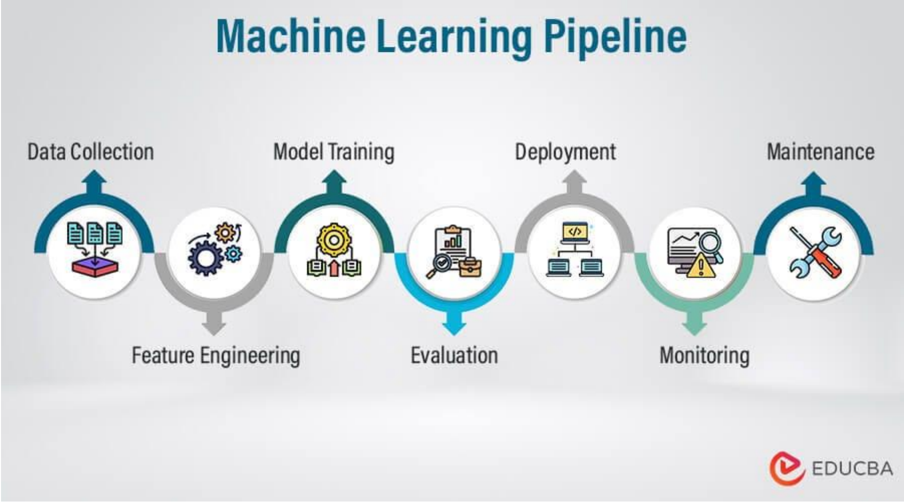
Supervised vs Unsupervised Learning



Supervised vs Unsupervised Learning

Aspect	Supervised	Unsupervised
Data	Labeled (X, y)	Unlabeled (X only)
Goal	Predict output	Discover patterns
Accuracy	Usually higher	Hard to measure
Effort	Labeling required	No labeling needed

Machine Learning Pipeline



tomate the process of building, training, and deploying ML

Steps to build Machine Learning Pipeline

Step 1: Data Collection and Preprocessing

- Gather data from sources like databases, APIs or CSV files.
- Clean the data by handling missing values, duplicates and errors.
- Normalize and standardize numerical values.
- Convert categorical variables into a machine readable format.

Step 2: Feature Engineering

- Select the most important features for better model performance.
- Create new features for feature extraction or transformation.

Step 3: Data splitting

- Divide the dataset into training, validation and testing sets.
- When dealing with imbalanced datasets, use random sampling.

Step 4: Model Selection & Training

- Choose the best algorithm based on the problem includes classification, regression, Clustering etc.
- Train the model using the training dataset.

Step 5: Model evaluation & Optimization

- Test the model's performance using accuracy, precision, recall and other metrics.
- Tune hyperparameters using Grid Search or Random Search and avoiding overfitting using techniques like cross-validation.

Step 6: Model Deployment

- Deploy the trained model using Flask, FastAPI, TensorFlow and cloud services.
- Save the trained model for real-world applications.

Step 7: Continuous learning & Monitoring

- Automates the pipeline using MLOps tools like MLflow or Kubeflow.
- Update the model with new data to maintain accuracy.

scikit-learn: Python ML Library

Built on NumPy, SciPy, and Matplotlib for production-ready ML

- **Installation:** pip install scikit-learn
- **Key modules:** preprocessing, model_selection, metrics, ensemble
- **Consistent API:** fit(), predict(), score()
- **Algorithms:** Classification, Regression, Clustering, Dimensionality Reduction

Classification Algorithms in scikit-learn

Simple Models

- Logistic Regression
- Naive Bayes
- SVM (Support Vector)ke KNN or regression

Complex Models

- Decision Trees
- Random Forest
- Gradient Boosting

Regression Algorithms in scikit-learn

Predicting continuous numerical values.

Linear Methods

- Linear Regression
- Ridge / Lasso
- ElasticNet

Non-Linear Methods

- SVR (Support Vector)
- Random Forest Regressor
- Gradient Boosting Regressor

Clustering Algorithms (Unsupervised)

Grouping similar data points without labeled data

- **K-Means:** Partitions data into K clusters based on centroids
- **DBSCAN:** Density-based clustering, handles arbitrary shapes
- **Hierarchical Clustering:** Creates tree-like cluster hierarchy
- **Gaussian Mixture Models:** Probabilistic clustering approach

Classification Evaluation Metrics

Assessing model performance and quality

Accuracy-Correct predictions / Total predictions

Precision-True Positives / (TP + False Positives)

Recall-True Positives / (TP + False Negatives)

F1-Score-Harmonic mean of Precision & Recall

Regression Evaluation Metrics

Measuring prediction accuracy for continuous values.

AMAE (Mean Absolute Error): Average absolute difference between predicted and actual

MSE (Mean Squared Error): Average squared differences (penalizes large errors)

RMSE (Root Mean Squared Error): Square root of MSE, interpretable unit

R^2 Score: Proportion of variance explained (0-1, higher is better)

Understanding Overfitting

Model memorizes training data, performs poorly on new data.

Causes

- Model too complex
- Too many features
- Limited training data
- Training for too long

Indicators

- High training accuracy
- Low test accuracy
- Gap increases with time
- Bad generalization

Preventing Overfitting: Techniques

Model memorizes training data, performs poorly on new data.

- **Regularization (L1/L2):** Penalize large weights to simplify model
- **Cross-Validation:** Evaluate on multiple data splits for reliable estimate
- **Early Stopping:** Stop training when validation performance plateaus
- **Dropout:** Randomly deactivate neurons during training (deep learning)
- **Data Augmentation:** Increase training data variety

Hyperparameter Tuning

Optimizing model parameters set before training

Key Hyperparameters

- Learning rate
- Number of hidden layers
- Regularization strength
- Tree depth (decision trees)

Search Methods

- Grid Search (exhaustive)
- Random Search (sampling)
- Bayesian Optimization
- Evolutionary algorithms

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